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DISCUSSION FORUM

ISO 5725 and GUM: comparison and comments

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Abstract We compare the approach to measure uncertainties proposed in ISO 5725 and GUM from a statistician point of view. In particular we give some warnings to the application of the expanded uncertainty introduced in GUM when the input variables are few and we report some considerations on the relevant role of the interactions among the input variables in the measurement equation as well as the role of statistical design of experiments to measure uncertainties.

Keywords Measurement equation · Confidence intervals · Design of experiments

Introduction

In the last 15 years the International Organization of Standardization (ISO) has produced several documents on uncertainty measurement. The interest of ISO in this topic relies on the consideration that quantitative evaluation of the uncertainty of a measurement is necessary to say whether the differences among the results of an experiment or the variability of a process are due to either an out of control process or to a not capable measurement system. This is relevant because process variation is becoming small due to technological improvements and therefore measurement procedures must be deeply tested to assess their adequateness (see, for a review [1]). The most important international standards on this topic are the ISO 5725:1994 (Part 1–6) [2] and the ISO "Guide to the expression of Uncertainty in Measurement" (GUM) [3] published in 1995, both proposing a method to calculate the uncertainty of a measurement.

In this paper we comment these two approaches mainly from a statistician point of view and at the light of a research in collaboration with a microelectronic company. In this field, it is easy to imagine that accuracy of measurement systems is essential due to the very small dimension of their products. Almost every day the capability of very expensive measurement instruments, used both in the production and in the development phase of a product, is tested, using, where appropriate, either the approach based on the gauge repeatability and reproducibility studies (GRR) (that is the principles in the ISO 5725) or the GUM procedure. The same issue may be of interest for the testing and calibration laboratories, as ISO/IEC/EN 17025 [4] standard demands that "Testing laboratories shall have and shall apply procedures for estimating uncertainty of measurement" and suggests to use ISO 5725 and GUM to compute measurement uncertainties.

To stress furthermore the relevance of this topic, a round table was recently organized on the evolution of these two standards, within the Advanced Mathematical and Computational Tools in Metrology and Testing Conference, held in Paris in June 2008.

We are aware that this is an area traditionally involving scientists from chemistry, physics, biochemistry and other related sciences, but recently it has been broadening, including statisticians too. In fact many arguments reported in the standards concern concepts and methods typical of

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this area, mostly due to the randomness that affects the measurement.

In particular in "ISO 5725 and GUM approaches to measure uncertainty" analogies and differences between ISO 5725 and GUM are illustrated. In "Some consideration about GUM" some warnings to the ISO users are given in order to avoid wrong applications of the standards and, consequently, wrong evaluation to measure uncertainty. In "How to collect data to define the uncertainty of a measurement?" we report some considerations on statistical experimental design in the computation of the uncertainty of a measurement.

ISO 5725 and GUM approaches to measure uncertainty

ISO 5725 and GUM approaches to the validation of the measurement procedure are quite different even if they were published almost in the same year. The main differences are about: (a) the terms used (e.g. accuracy in ISO 5725, uncertainty in GUM); (b) how to gather data in the measurement procedure; (c) the model and the techniques to analyze data; (d) how to present the results. We will analyze these differences in "Comparison between ISO 5725 and GUM", after having introduced the main characteristics of both approaches in "ISO 5725: measurement methods and results" and "The guide to the expression of uncertainty in measurement (GUM)".

ISO 5725: measurement methods and results

ISO 5725 deals with accuracy of measurement methods and results. It uses two terms—*trueness* and *precision*—to describe the accuracy of a measurement method. Trueness refers to the closeness of agreement between the arithmetic mean of a large number of test results and the accepted reference value, while precision refers to the closeness of agreement between test results. Two measure of trueness may be of interest: bias of the measurement method (the difference between the expectations of test results obtained from all laboratories using that method and the accepted reference value) and the laboratory bias (the difference between the expectation of the test results for a particular laboratory and the accepted reference value). In both cases a reference value is required. On the other hand, precision does not require a reference value, as it involves only comparisons between replicate measurements of a quantity under specified conditions. It can be measured under repeatability or reproducibility conditions. The former exists when independent test results are obtained with the *same* method on identical test items in the *same* laboratory by the *same* operator using the *same* equipment within short time intervals. Reproducibility conditions exist when test results are obtained with the *same* method on identical test items in *different* laboratories with *different* operators using *different* equipment. Thus repeatability and reproducibility are the two extremes of precision, the former describing the minimum and the latter the maximum variability of the results.

Then ISO 5725 concerns exclusively with measurement methods which yield measurements y typically on a continuous scale with a single value for each test result. The y values describe, in quantitative terms, the ability of a method to give a correct result (trueness) or to replicate a given result (precision) and they are obtained by a collaborative study, that is an interlaboratory experiment run under reproducibility condition.

The model explaining the measurand Y depends on the experimental design used to obtain the measurement. It is generally a random effect model such as

$$Y = \mu + B + E \tag{1}$$

where μ is the overall mean, B is the laboratory component of bias under repeatability conditions and it is assumed to be normally distributed as $N(0, \sigma_L^2)$, E is the random error under repeatability conditions assumed to be $N(0, \sigma_E^2)$, B and E are supposed to be uncorrelated.[1] Observe that σ_L^2 includes the between-operator and the between-equipment variability, while σ_E^2 is the repeatability variance. The reproducibility standard deviation $\sigma_R = \sqrt{\sigma_L^2 + \sigma_E^2}$ is estimated using the analysis of variance (ANOVA) method.

In ISO 5725 no confidence interval for μ is present. Only repeatability (r) and reproducibility (R) limits are defined. These limits are the values less than or equal to which the absolute difference between two test results, obtained under either repeatability or reproducibility conditions, may be expected to be with $(1 - \alpha)$ probability level. Given model in Eq. 1, r and R corresponds to

$$r = t_{df,\alpha/2} \cdot s_r \cdot \sqrt{2}, R \cong t_{df,\alpha/2} \cdot s_R \cdot \sqrt{2}$$

where $t_{df},\alpha/2$ is the upper $\alpha/2$ percentile of a t Student distribution, df are appropriate degrees of freedom and s_r, s_R are the square roots of the estimate of the variance under repeatability and reproducibility conditions, respectively. Observe that R is defined on the basis of an approximation because s_R^2 corresponds to a linear combination (with positive signs) of chi-squared random variables for which an exact distribution does not exist.

[1] The assumption of uncorrelation between B and E is not present in ISO 5725, but it is correctly reported in the Appendix A.2 "Collaborative study approach" of the ISO 21748:2004 [5].

Table 1 Comparison between ISO 5725 and GUM

Scope	ISO 5725 Accuracy: trueness and precision	GUM Measurement uncertainties
Measurand	Y is measurable	Y may be not directly measured
Data obtained by	Statistical experimental design	NO statistical experimental design
Model	$Y = \mu + B + E$	$Y = f(X_1, X_2,\ldots,X_m)$
Variability of the measurand	Repeatibility and reproducibility variance	Combined standard uncertainties $u_c^2(y)$
Confidence interval	NO: only the definition of repeatibility and reproducibility limit	NO: only the definition of an interval, $y \pm U_p$, that could reasonably be attributed to the measurand.

The guide to the expression of uncertainty in measurement (GUM)

On the contrary to the ISO 5725, in the GUM framework, Y may not be directly measured and then no experimental design is generally run to obtain data. The basic idea is that a measurement equation exists. It may be a known (or even an unknown) function generically written as follows:

$$Y = f(X;\theta) \quad X \in \mathrm{B} \subset \mathrm{R}^m \tag{2}$$

where $\boldsymbol{X} = (X_1,\ldots,X_m)$ is a vector of m random variables and $\boldsymbol{\theta} = (\theta_0, \theta_1,\ldots,\theta_m)$ is a vector of $m + 1$ coefficients. Given a vector of values, $\boldsymbol{x} = (x_1,\ldots,x_m)$, the estimate y is obtained from Eq. 2 as

$$y = f(x;\boldsymbol{\theta}) \tag{3}$$

where $(x_1,\ldots,x_m)$ may be composed of either/both the mean of n independent repeated measurements of $\boldsymbol{X}$ (called Type A method) or/and the expected mean of the probability density function (PDF) assumed for $\boldsymbol{X}$ (called Type B method). The latter may be based also on non statistical evaluations including expert judgment.

To avoid the non linearities in Eq. 2 the Taylor series expansion of $f(\boldsymbol{X};\boldsymbol{\theta})$ is calculated in the neighbourhood of $\boldsymbol{X} = \boldsymbol{x}$. Truncating the expansion at the first order, we may approximate Eq. 2 as follows

$$Y \approx c_0 + \sum c_i X_i \tag{4}$$

where $c_i = [\partial f/\partial X_i]_{X_i = x_i}$ $(i = 1,\ldots,m)$ are the "sensitivity coefficients". Putting $\boldsymbol{x}$ in Eq. 4 we obtain an approximation of the unknown value of y in Eq. 3. As it is often necessary to find the lower and the upper bound for y, GUM introduced the concept of expanded uncertainty, U_p. It is defined as $U_p = k \cdot u_c(y)$, where

$$u_c(y) = \sqrt{\sum c_i^2 u^2(x_i)} \tag{5}$$

is called combined uncertainty of y. It depends on the uncertainties $u(x_i)$ of each input variables and k is a constant called coverage factor. Formula 5 is based on the assumption that the input variables are not correlated.

Around the measurement y the expanded uncertainty, U_p, defines an interval $[y \pm U_p]$ that should include the values that could reasonably be attributed to the measurand. The coverage probability of the interval $[y \pm k\, u_c(y)]$ is generally not directly measurable since a state-of-knowledge probability distribution for Y cannot be exactly determined in general. Relying on the central limit theorem, the probability density function (PDF) of Y is in many cases supposed to be Gaussian. $k = 2$, or an appropriate percentile of the t Student, is generally used to guarantee that the confidence interval has an expected confidence probability level of 0.95.[2]

From the beginning GUM was a widely adopted standard approach. It gave the possibility to take into consideration the uncertainty of Type B, that is uncertainties specified by information not obtained through collaborative trials, such as calibration. In the practice the experimenter has to prepare "only" the "uncertainty budget", that is the list of all input variables X_i in Eq. 2 with their corresponding $u(x_i)$ and c_i.

Comparison between ISO 5725 and GUM

As it is clear from the previous paragraphs there are many differences between the two ISO guidances. A comparison is reported in Table 1.

We observe that:

(a) ISO 5725 and GUM use different terms but they share the same goal, as uncertainty of a measurement is the opposite of its accuracy. In fact VIM defines *accuracy* of the measurand as "the closeness of agreement between a quantity value obtained by measurement and an accepted reference value", while *uncertainty* is defined in GUM (clause 2.2.3) as "a parameter associated with the result of a measurement that characterizes the dispersion of the values that could

[2] Note that k should be 1.96 to guarantee exactly 0.95 if Y is supposed to be normally distributed.

reasonably be attributed to the measurand". It follows that the greater is the dispersion of the values the greater is the uncertainty of the measurement and the lower is its accuracy.

(b) In ISO 5725 the measurand is directly measurable on the contrary to GUM, where the measurand may be not directly measured, for example when the measurand is an area or a volume.

(c) Data are obtained by a statistical experimental design in ISO 5725 and not so in GUM: the consequences of this approach will be discussed in "How to collect data to define the uncertainty of a measurement?".

(d) Eq. 1, explaining the measurand in the ISO 5725, is a statistical model, while $f(\boldsymbol{X};\ \boldsymbol{\theta})$, in the GUM framework, is a mathematical function.

(e) The combined standard uncertainties $u_c(y)$ is computed in GUM through the law of propagation of uncertainties, while s_r and s_R are computed in ISO 5725 by ANOVA method.

(f) GUM defines an interval around the measurement y while in ISO 5725 only the repeatibility and reproducibility limits (r, R) are defined.

However, analogously to GUM, in the ISO framework an interval around y may be found. In fact, under reproducibility conditions, we may assume that

$$\left[y \pm t_{df,\alpha/2} \cdot s_R/\sqrt{n}\right]$$

corresponds to an approximate $(1 - \alpha)$ confidence interval for $\mu = E[Y]$. The interval is an approximation because, as already mentioned in "ISO 5725: measurement methods and results", s_R^2 is approximately χ^2 distributed with degree of freedom, df, computed using the Welch–Satterwhaite formula.

It is clear that the GUM approach, unlike ISO 5725, allows to consider also the calibration uncertainties, so guaranteeing the traceability of the measurement instruments to primary standards [6].

For some particular circumstances (linearity of the function $f(\boldsymbol{X};\boldsymbol{\theta})$ in Eq. 2, normality distribution and independence of the input variables) the estimates of the variance of the measurand obtained by the two approaches may be comparable. This is probably the reason these two guidances are sometimes proposed to be alternative (as in [4]) or even to be used jointly.

A combination of the ISO 5725 and GUM's approaches may be found in two ISO technical specifications (ISO/TS 21748:2004 [5], ISO/TS 21749:2005 [7]).

ISO 21749 follows the GUM approach to compute the expanded uncertainties but it proposes to use the analysis of variance to estimate uncertainties of components classified as Type A. This guidance considers nested designs to estimate the uncertainties due to time-dependent sources of variability, to measurement configuration (differences due to different instruments, operators, geometries) or to material inhomogeneity, while it uses the GUM approach to compute the expanded uncertainty.

ISO 21748 deals with the evaluation of measurement uncertainties using data obtained from studies run in accordance with ISO 5725-2, that is relative to the inter-laboratory experiments. These studies yield performance indicators (s_r and s_R) estimated using Eq. 1. In the practice s_R does not necessarily include variation of all the effects that influence a measurement result. This is the reason for which this guidance modifies Eq. 1 introducing the following model

$$Y = \mu + \delta + B + \sum_{i=1}^{m} c_i X_i^* + E \qquad (6)$$

where in particular δ represents the intrinsic bias due to the measurement method, X_i^* are input variables assumed to be $N(0, u^2(x_i))$ (in the GUM approach of type A only) and the summation is over all those effects subject to deviations other than those incorporated in B or E. Then Eq. 6 is a combination of Eq. 1 used by ISO 5725 and Eq. 4 used by the GUM approach.

Some consideration about GUM

The attractive features of the GUM framework contrast with the many assumptions that are usually necessary to find U_p: the application of the law of propagating errors, the distributional assumption especially for Type B components, how to choose k (we remind the many criticism appeared in the metrological literature, see, e.g. the special issue of *Metrologia* in 2006), the assumption of independence (or, less strongly, of uncorrelation) among $(X_1,\ldots,X_m)$ and so on. In particular the first order Taylor expansion in Eq. 4 may be a poor approximation if significant non linearities exist. We agree with [8] when it is said that every method of evaluation of expanded uncertainty is an approximate method, and the degree of approximation cannot be evaluated because there is often no defined reference method that is considered accurate. Therefore, if we wish to discuss the accuracy of approximate methods of evaluation of expanded uncertainty, we must define an accurate method against which the approximate methods may be compared.

In this direction [9] constitutes a very significant improvement. Conditions for valid application of GUM are there reported in detail, as well as how to apply Monte Carlo Methods (MCM) (see also, e.g. [10]) to compute (generate) the distribution of Y in Eq. 2 is broadly commented.

Maybe the major property of MCM methods is that an interval for y, e.g. $[y_{low}, y_{high}]$, such that it almost exactly

encompasses a desired $(1 - \alpha)$ fraction of the simulated distribution for Y instead of the approximate interval $[y \pm U_p]$ may be found. In particular, MCM works well even when the distribution of Y is asymmetric. Analogously to the GUM approach, MCM cannot manage dependencies among $\boldsymbol{X}$ if no prior information are formulated or if no appropriate design of experiment is run to measure the correlations. May be a suggestion is to put a prior also on the correlation matrix among $\boldsymbol{X}$ or to adopt the opinion of an expert or even to exploit the power of the so-called *copulas*. But all these solutions are far to be practical and certainly they contrast with the main property of GUM: the one to be a quite easy technique to manage Eq. 2. Some hints are available in [9] on how to deal with these not irrelevant details. Many other detailed technical notes are in the Supplement 1 to the GUM and interesting is, among the many issues, the proposal to use the principle of maximum entropy to choose an appropriate distribution for $\boldsymbol{X}$ when no experimental data are available.

Notwithstanding the simplicity of the MCM approach we believe that if Eq. 2 is not too complex or the number of input variables is small and for a matter of simplicity (Eq. 4) is thought of simply as a sum (or a weighted sum) of variables, it is better to apply the method of transformation of random variables because, given the experimenter's assumptions, we may find mathematically the interval that encompasses the desired $(1 - \alpha)$ fraction of the exact distribution for Y. Such a little bit more complex approach may solve both the computational time requested by the MCM approach and the validation of the procedure. In addition it is possible to show that the convolution of (few) random variables may be numerically done even with very easy tools such as a spreadsheet. The major drawback is the possibility to appreciate how much great the approximations are due to the adoption of the expanded uncertainty, U_p, with respect to the exact solution, even for a very wide range of alternatives for the domain of $\boldsymbol{X}$.

To show this, we may say that given Eq. 2, supposing that the joint PDF of $\boldsymbol{X}$ is known and that the inverse f^{-1} exists at least on a partition of the domain, then we may find the distribution for Y as follows: suppose $g_X(\boldsymbol{x}; \boldsymbol{\lambda})$ is the (multivariate) PDF of $\boldsymbol{X}$ and let arbitrarily $Y_1 \equiv Y = f(\boldsymbol{X};\boldsymbol{\theta})$. Using f^{-1}, let

$$f^{-1}(Y_1, X_2, \ldots, X_m; \theta) = X_1.$$

To find the PDF of Y the method of transformations of random variables may be used. We must introduce $(m - 1)$ 1:1 transformations such that $Y_i \equiv X_i$ for $i = 2,\ldots,m$.

Call $\mathbf{J}$ the Jacobian matrix with elements (i,j) equal to $\partial X_j/\partial Y_i$ for $i, j = 1,2,\ldots,m$. Then the PDF of $g_Y(y; \boldsymbol{\lambda}, \boldsymbol{\theta})$ will be given by

$$g_Y(y;\boldsymbol{\lambda},\boldsymbol{\theta}) = \int \ldots \int |\mathbf{J}|\, g_{\mathbf{X}}(y_1, x_2, \ldots, x_m; \boldsymbol{\lambda}) dx_2 \cdots dx_m \tag{7}$$

where $|\mathbf{J}| = \partial f^{-1}(y, x_2,\ldots,x_m; \boldsymbol{\theta})/\partial y$.

Supposing, as it is often assumed by GUM and in the practice, that the variables in $\boldsymbol{X}$ are independent and letting, without loss of generality and for a matter of simplicity, $c_0 = 0$ and $c_i = 1\ \forall i$ in Eq. 4, Y is a convolution (less properly it may be thought of as a sum) of random variables. Then Eq. 7 becomes

$$g_{\boldsymbol{Y}}(y;\boldsymbol{\lambda}) = \int \ldots \int g_{\mathbf{X}_1}(y_1 - x_2 - \cdots - x_m; \boldsymbol{\lambda}_{\mathbf{X}_1}) \times g_{\mathbf{X}_2}(x_2; \boldsymbol{\lambda}_{\mathbf{X}_2}) \cdots g_{\mathbf{X}_m}(x_m; \boldsymbol{\lambda}_{\mathbf{X}_m}) dx_2 \cdots dx_m \tag{8}$$

which may be solved iteratively, considering first the convolution of $X_1 + X_2 = Y_{12}^*$, then the convolution of $Y_{12}^* + X_3 = Y_{123}^*$ and so forth.

To exemplify, Eq. 8 has been implemented in Excel for a case that is very common in the practice. We have created in a single cell of the spreadsheet a formula that computes the probability density function of Eq. 8 supposing $X_1 \sim N(0, \sigma^2)$ and $X_i \sim U(-a_i, +a_i)$ for $i = 2,\ldots,5$. This case corresponds to the hypothesis that X_1 is a Type A variable and the others are Type B uniform random variables (with different domain). Of course the same example may be modified substituting the gaussian distribution with another one. We have chosen the gaussian distribution just for a matter of simplicity well knowing that it is often an idealization of reality. It may be thought of as the application of the central limit theorem when many Type A input variables are considered. Additionally, we make notice that also in [9] with the exception of only two cases, most of the suggested distributions for $\boldsymbol{X}$ are symmetric, and then the convergence of Eq. 4 to normality should be sufficiently fast.

Suppose that $[y \pm k u_c(y)]$ is found using $k = \Phi^{-1}(0.975) = 1.96$, where $\Phi(\cdot)$ is the cumulative density function (CDF) of a $N(0,1)$ random variable.

Figure 1 is the scatter plot of $v = \sqrt{\frac{Var(X_2+X_3+X_4+X_5)}{Var(X_1)}}$ for $\sigma^2 = 0.2(0.2)2$ and $a_i = 0.2(0.2)2$ versus the ratio $r = \frac{Pr^{-1}(Y \le 0.975)}{\Phi^{-1}(0.975)\sigma_U}$ where $Pr^{-1}(Y \le 0.975) = y_{0.975}$ is the percentile of order 0.975 of the convolution Y and $\sigma_U = \sqrt{\frac{a_2^2}{3} + \frac{a_3^2}{3} + \frac{a_4^2}{3} + \frac{a_5^2}{3} + \sigma^2}$.

From Fig. 1 it is clear that as the contribution to the overall variability due to the uniform distributions increases, that is as v increases, r decreases, showing that having fixed to 0.95 the $(1 - \alpha)$ probability level of the interval $[y \pm k\ u_c(y)]$, the practitioners choice of $k = \Phi^{-1}(0.975)$ overestimates the exact value found using the convolution.

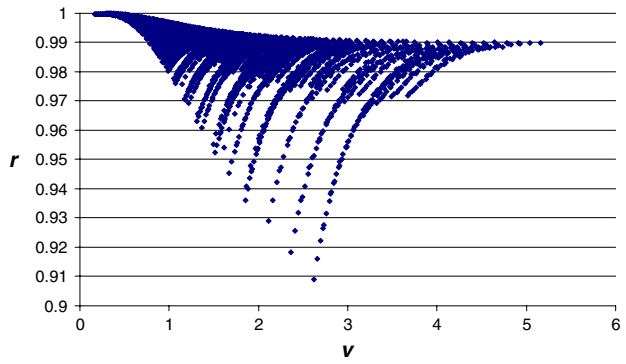


Fig. 1 Scatterplot of the approximations due to the adoption of the expanded uncertainty, U_p, with respect to the exact solution found using the convolution of the random variables for the example in "Some consideration about GUM" [X axis : $v = \sqrt{\frac{Var(X_2+X_3+X_4+X_5)}{Var(X_1)}}$; Y axis : $r = \frac{Pr^{-1}(Y \leq 0.975)}{\Phi^{-1}(0.975)\sigma_U}$]

The major drawback is that the interval $[y \pm U_p]$ is wider than necessary so that the probability to consider a measurement within the uncertainty limits when it is not, increases. The minimum of r has been observed at $\sigma^2 = 0.2, a_2 = 2, a_3 = a_4 = a_5 = 0.2$, confirming that if a dominant contribution to the combined uncertainty from an input quantity exists, the choice of $k = \Phi^{-1}(0.975)$ is far from the true value.

Certainly using the GUM approach, how to include the assumption of dependence among $(X_1,\ldots,X_m)$ continue to be a problem. The reason may be addressed to the fact that using the GUM, the experimenter samples from X_i marginally with respect to all the other variables and this does not allow to evaluate the dependence among the variables. The consequence is that $u_c(y)$ may be over or under estimated depending on the sign of the covariances if they are not zero.

May be the previous issue is not a problem at all for the practitioners. Those who apply the GUM procedure are interested in completing the uncertainty budget [11] without any validation of the assumption of independence among input quantities (see, e.g. [12]). Analogously no explicit function (Eq. 2) may be formulated resulting the combined uncertainty to be equal to Eq. 5 (see, e.g. [13]).

How to collect data to define the uncertainty of a measurement?

How to gather data to define the uncertainty (or accuracy) of a measurement is an important task. ISO 5725 and GUM seems to propose a quite different answer to this question.

The use of experimental designs (DOE) (see, e.g. [14, 15]) to obtain data is a natural procedure in ISO 5725 where the relevance of DOE is also underlined by the request of the presence, in the panel of experts, of at least a person with experience in statistical design and analysis of experiment and familiar with the measurement method and its application (e.g. see clause 6.1.1 of ISO 5725: Part 1).

Using a different approach, in GUM n independent repeated measurements of input quantities are requested to estimate the uncertainties of type A while only expert judgement and/or probability distributional assumptions are necessary to estimate the uncertainties of type B. Some interesting considerations on repeated measurements in the GUM context in contrast with replicated measurements are reported in [16].

Even if, generally, in ISO 5725 simple design (one factor or nested design) are considered, the opportunity to profitably extend the use of DOE to collect data in order to measure uncertainty is underlined, e.g. in [11], to include design with several factors (factorial, nested or split-plot) and mixed design (with fixed and/or random factors). In [17] a Central Composite Design (CCD) and a correspondent response surface model for the experimental estimation of measurement uncertainty in chemical measurement is proposed.

Depending on the kind of experiment and factors, i.e. depending on how data are collected, an appropriate model to explain the measurand can be identified and statistical methods are available to estimate the parameters of interest and variance components using ANOVA. Additionally the main by-products of obtaining data by a statistical experiment are:

1. the randomization of the trials in the design guarantees the independence of the random effects and the error component;
2. the factorial experimental design gives the possibility to estimate, if they exist, interactions among factors (the latter is in particular not possible in the GUM procedure if not explicitly requested);
3. numerous statistical results are available to obtain approximated confidence intervals of variance components as modified large sample (MLS), generalized confidence intervals (GCI) and simulated confidence intervals (SCI). These methods generally performs better than the Welch–Satterwhaite approximation [18, pp. 146–147] as it allows to find more precise confidence intervals and they could easily be used in metrological context to assess whether each variance component is relevant before pooling them to obtain the estimate of the variance of Y.

Can experimental design be useful to collect data also in GUM approach?

As it has been suggested by a referee—and we agree with him—when the measurement is measurable, DOE should be profitably adopted to collect data, both to

establish which input quantities in (2) are relevant to explain the uncertainties of the measurand and to estimate the uncertainties components of the input quantities of type A. In fact DOE allows to collect data and analyze them by statistical methods using the smallest number of trials and then at the minimum cost (see, e.g. [19]), and, most important, to isolate the effect of each factor to determine if their interactions are relevant. In many parts of GUM and in [9], it is written that possible dependence among input quantities have to be taken into account. But it is not absolutely explained to the users how to check such a dependence (see, e.g. [20]). DOE can resolve this question as it allows to put in evidence the effective relevance of each input quantities on Y and their possible correlation both in the case of random and fixed factors. Sometimes the dependences among input quantities are due to the effect of a dominant variable and, if data are collected marginally, these effects are masked. In [9] it is said that "the evaluation of uncertainty is neither a routine task nor a purely mathematical one; it depends on detailed knowledge of the nature of the measurand and of the measurement method and procedure used. The quality and utility of the uncertainty quoted for the result of a measurement therefore it ultimately depends on the understanding, critical analysis, and integrity of those who contribute to the assignment of its value". In this sense DOE can represent a substantial support to know the nature of the measurand and to understand what are the sources of uncertainties to be taken into account to estimate the global uncertainties (see, e.g. [21]).

Also in the case in which the measurement is not measurable but the function f in Eq. 2 and the input quantities (of type A) are known, DOE can help to collect data in a way suitable to show the possible relations among dependent variables. Consider the simple case: Y is the area of a rectangular ($Y = L \times W$) and follow the GUM approach to estimate its standard deviation. We compute separately the standard uncertainties associated with L, W and combine the two standard uncertainties into one, using the approximation for the product of two variables. If we adopt an appropriate DOE to collect data on L and W of the rectangular we may take into account also the influence that, for example, the environmental condition (temperature, humidity,…) can simultaneously have on L and W and in this way we can better estimate the uncertainties of Y.

Conclusion

Measurement uncertainty is a vital issue within analytical science. The question is: what is the most appropriate method to estimate the measurement uncertainty? Generally speaking we may compare a modelling method (GUM) and an empirical method (based on an experimental design as ISO 5725). In modelling, the estimation of uncertainty involves the identification, quantification and summation (as variances) of each potential source of uncertainty. This approach becomes increasingly problematic in identifying all of such sources when it is applied, e.g. to primary sampling. On the other hand the empirical approach to uncertainty estimation involves replicated measurements from either inter-laboratory trials and/or internal method validation and quality control. In [22] a strategy for reconciling the two approaches is proposed. In [23] the strengths and weakness of the two methods of uncertainty estimation are discussed and it has been proved, by a case study, that the estimate of sampling uncertainty made using the modelling approach resulted to be six times larger than that found using the empirical approach. The difficulty in establishing reliable estimates for the input variable for the modelling approach is thought to be the main cause of the discrepancy and the empirical approach to uncertainty estimation was recognized to be generally the one providing the more reliable estimates. If the response variable Y is measurable and the function that links it to the independent variables is not known, the design of a suitable statistical experiment and the use of the correspondent ANOVA table provide the best approach to estimate the different sources of variation derived from a measurement system.

But when Y is not measurable or there are input variables of type B, the GUM approach to obtain the expanded uncertainty should to be used choosing with care the value of k and being aware the assumption of independence among factors may be relevant. GUM on one hand is apparently easier to be implemented and with the introduction of the type B variables it gives the users great flexibility to the choice of the kind of input variables, but, on the other hand, it gives them also the responsibility of the exactness of the results. In our experience, even when Y is measurable we have noticed a liberal approach towards the implementation of the GUM especially in the definition of Eq. 2 (often considered as a sum of input variables) and no preliminary check for the presence of correlation among variables is usually made. In this context DOE could help practitioners in collecting data to measure uncertainties even using GUM procedure.

Finally, even if in ISO standards the uncertainty due to the manufacturing process is never taken into account in the model as "measurement of check standards are used", in the practice not always the measurements are referred to a check standard. In these cases it is necessary to introduce the factor "part" in the model and to consider also its interaction with the other factors. The use of mixed effect model is then an appropriate solution that may help to consider jointly fixed and random factors in a single

factorial (crossed or nested) experiment to estimate all uncertainties components (systematic and random).

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